Classify an sjSDM Unit Pipeline Error

Description

Distinguishes registered data-feasibility outcomes from unexpected pipeline failures using target-level error metadata.

Usage

```
classify_sjsdm_unit_pipeline_error(data_target_errors = NULL)
```

Arguments

`data_target_errors`	Data frame containing at least 'name' and 'error' columns for errors recorded during one unit execution.

Details

Expected roots must match both a registered target-name pattern and a stable message pattern. Downstream dependency, missing-object, and empty-pattern errors are allowed only as cascades. Known temporal validation cascades are accepted only when the same time slice has a registered expected root. Any other root remains unexpected. A missing-qs reader fallback is a cascade only beside a registered data root. Alone it remains unexplained and must never authorize an exclusion. Counted and legacy insufficient-row MEM errors are registered only at the shared basis target; arbitrary MEM engine failures remain unexpected. Verified zero-positive-eigenvector support is recorded separately as 'no_positive_spatial_eigenvectors', requiring at least one positive vector.

Value

A list with 'status', 'reason_code', 'observed_count', 'required_count', 'root_target', and 'root_error'.